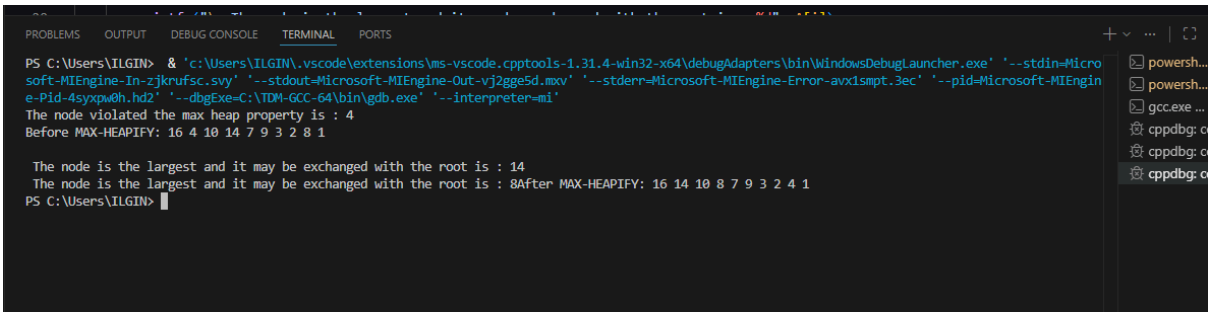
Algorithms and Data Structures

REPORT LAB 5

Name: ILGIN
Surname: CEYLAN
Student ID: 82862

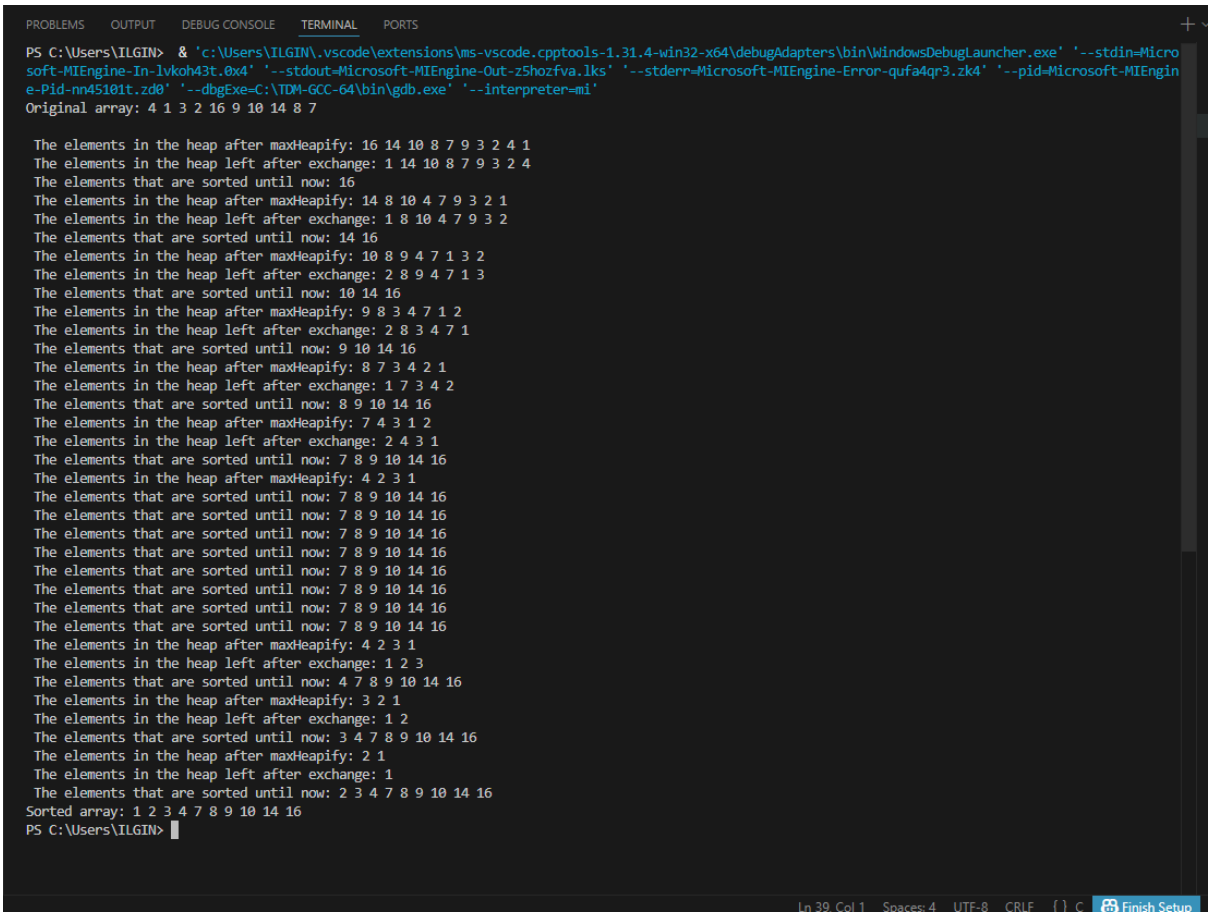
TASK1


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\ILGIN> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-zjkrufsc.svy' '--stdout=Microsoft-MIEngine-Out-vj2gge5d.mxv' '--stderr=Microsoft-MIEngine-Error-avx1smpt.3ec' '--pid=Microsoft-MIEngine-Pid-4syxpw0h.hd2' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
The node violated the max heap property is : 4
Before MAX-HEAPIFY: 16 4 10 14 7 9 3 2 8 1
The node is the largest and it may be exchanged with the root is : 14
The node is the largest and it may be exchanged with the root is : 8After MAX-HEAPIFY: 16 14 10 8 7 9 3 2 4 1
PS C:\Users\ILGIN>
powersh...
powersh...
gcc.exe ...
cppdbg:
cppdbg:
cppdbg:

TASK2


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\ILGIN> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-crw1ghlb.wsc' '--stdout=Microsoft-MIEngine-Out-ayndag5i.o3g' '--stderr=Microsoft-MIEngine-Error-2nicgn3h.vtw' '--pid=Microsoft-MIEngine-Pid-zs1gjruk.b2p' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
Orijinal array: 4 1 3 2 16 9 10 14 8 7
The root that we run maxHeapify on it is 16
The root that we run maxHeapify on it is 2
The root that we run maxHeapify on it is 3
The root that we run maxHeapify on it is 1
The root that we run maxHeapify on it is 4
Max-Heap array: 16 14 10 8 7 9 3 2 4 1
PS C:\Users\ILGIN>

TASK3


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\ILGIN> & 'c:\Users\ILGIN\.vscode\extensions\ms-vscode.cpptools-1.31.4-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe' '--stdin=Microsoft-MIEngine-In-lvkoh43t.0x4' '--stdout=Microsoft-MIEngine-Out-z5hozfva.lks' '--stderr=Microsoft-MIEngine-Error-qufa4qr3.zk4' '--pid=Microsoft-MIEngine-Pid-nn45101t.zd0' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '--interpreter=mi'
Original array: 4 1 3 2 16 9 10 14 8 7
The elements in the heap after maxHeapify: 16 14 10 8 7 9 3 2 4 1
The elements in the heap left after exchange: 1 14 10 8 7 9 3 2 4
The elements that are sorted until now: 16
The elements in the heap after maxHeapify: 14 8 10 4 7 9 3 2 1
The elements in the heap left after exchange: 1 8 10 4 7 9 3 2
The elements that are sorted until now: 14 16
The elements in the heap after maxHeapify: 10 8 9 4 7 1 3 2
The elements in the heap left after exchange: 2 8 9 4 7 1 3
The elements that are sorted until now: 10 14 16
The elements in the heap after maxHeapify: 9 8 3 4 7 1 2
The elements in the heap left after exchange: 2 8 3 4 7 1
The elements that are sorted until now: 9 10 14 16
The elements in the heap after maxHeapify: 8 7 3 4 2 1
The elements in the heap left after exchange: 1 7 3 4 2
The elements that are sorted until now: 8 9 10 14 16
The elements in the heap after maxHeapify: 7 4 3 1 2
The elements in the heap left after exchange: 2 4 3 1
The elements that are sorted until now: 7 8 9 10 14 16
The elements in the heap after maxHeapify: 4 2 3 1
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements that are sorted until now: 7 8 9 10 14 16
The elements in the heap after maxHeapify: 4 2 3 1
The elements in the heap left after exchange: 1 2 3
The elements that are sorted until now: 4 7 8 9 10 14 16
The elements in the heap after maxHeapify: 3 2 1
The elements in the heap left after exchange: 1 2
The elements that are sorted until now: 3 4 7 8 9 10 14 16
The elements in the heap after maxHeapify: 2 1
The elements in the heap left after exchange: 1
The elements that are sorted until now: 2 3 4 7 8 9 10 14 16
Sorted array: 1 2 3 4 7 8 9 10 14 16
PS C:\Users\ILGIN>
Ln 39, Col 1
Spaces: 4
UTF-8
CRLF
{ } C
Finish Setup